

1 Autocorrelation of Morlet wavelet

$$1. \quad \Sigma_{ac} = \lim_{T \rightarrow \infty} \int_{-T}^T |\psi(t)|^2 dt = \int_{-\infty}^{\infty} (e^{-t^2} \cos(\pi f_0 t))^2 dt$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} e^{-t^2} dt + \frac{1}{2} \int_{-\infty}^{\infty} e^{-t^2} \cos(2\pi f_0 t) dt = I_1 + I_2$$

$$I_1 = \frac{1}{2} I = \frac{\sqrt{\pi}}{2}$$

Gaussian integral:

$$I = \int_{-\infty}^{\infty} e^{-at^2} dt = \sqrt{\frac{\pi}{a}}$$

$$I_2 = \frac{1}{2} \int_{-\infty}^{\infty} e^{-t^2} \operatorname{Re}\{e^{2\pi i f_0 t}\} dt$$

$$= \frac{1}{2} \operatorname{Re}\left\{ \int_{-\infty}^{\infty} e^{-t^2} e^{2\pi i f_0 t} dt \right\} = \frac{1}{2} \operatorname{Re}\left\{ \int_{-\infty}^{\infty} e^{-(t - \pi i f_0)^2} e^{-\pi^2 f_0^2} dt \right\}$$

$$= \frac{1}{2} e^{-\pi^2 f_0^2} \operatorname{Re}\left\{ \int_{-\infty}^{\infty} e^{-(t - \pi i f_0)^2} dt \right\} = \frac{\sqrt{\pi}}{2} e^{-\pi^2 f_0^2}$$

Gaussian integral

$$\rightarrow \Sigma_{ac} = I_1 + I_2 = \sqrt{\frac{\pi}{2}} (1 + e^{-\pi^2 f_0^2})$$

$$2. \quad F\{e^{-t^2} \cos(\pi f_0 t)\} = F\{e^{-t^2}\} * F\{\cos(\pi f_0 t)\}$$

$$F\{e^{-t^2}\} = \int_{-\infty}^{\infty} e^{-t^2} e^{-2\pi i f t} dt = \int_{-\infty}^{\infty} e^{-(t + \pi i f)^2} e^{-\pi^2 f^2} dt$$

$$= e^{-\pi^2 \tau^2} \int_{-\infty}^{\infty} e^{-(t + \pi t j)^2} dt = \sqrt{\frac{\pi}{\tau}} e^{-\pi^2 \tau^2}$$

↑
Gaussian integral

$$F\{\cos(\sqrt{\pi} t_0 t)\} = \frac{1}{\sqrt{\pi}} (\delta(t - t_0) + \delta(t + t_0))$$

$$\rightarrow F\{e^{-t^2} \cos(\sqrt{\pi} t_0 t)\} = \sqrt{\frac{\pi}{\tau}} e^{-\pi^2 \tau^2} * (\delta(t - t_0) + \delta(t + t_0))$$

$$= \frac{\sqrt{\pi}}{\tau} \left(e^{-\pi^2 (t - t_0)^2} + e^{-\pi^2 (t + t_0)^2} \right)$$

3. if $x(t)$ is energy signal: $S_x(f) = F\{x(t) * x^*(-t)\}$

$$= |X(f)|^2 = \frac{\pi}{\tau} \left(e^{-\pi^2 \tau^2 (f - f_0)^2} + e^{-\pi^2 \tau^2 (f + f_0)^2} \right)$$

4. $S_x = F\{R_x(\tau)\} \rightarrow R_x(\tau) = F^{-1}\{S_x(f)\}$

$$= \int_{-\infty}^{\infty} S_x(f) e^{j\pi f \tau} df = \frac{\pi}{\tau} \frac{1}{\sqrt{\pi}} e^{-\frac{\tau^2}{\tau}} \left(e^{j\pi f_0 \tau} + e^{-j\pi f_0 \tau} \right)$$

$$= \frac{\sqrt{\pi}}{\tau \sqrt{\pi}} \left(e^{-\pi^2 \tau^2 f_0^2} + \cos(\pi f_0 \tau) \right)$$

2. Energy spectral density

$$1. E = \int_{-\infty}^{\infty} p(t) dt, \quad p(t) = \frac{v^2(t)}{R} \rightarrow E = \int_{-\infty}^{\infty} \frac{A^2 + u^2(t) e^{-\gamma \alpha t}}{R} dt$$

$$= \frac{A^2}{R} \int_0^{\infty} \underbrace{t e^{-\gamma \alpha t}}_I dt, \quad I = \frac{t e^{-\gamma \alpha t}}{-\gamma \alpha} \Big|_0^{\infty} - \int_0^{\infty} t e^{-\gamma \alpha t}$$

$$= \frac{A^2}{R \alpha^2}$$

$$2. F\{t e^{-\alpha t}\} = j \frac{d}{d\omega} F\{e^{-\alpha t}\} = j \frac{d}{d\omega} \frac{1}{\alpha + j\omega} = \frac{1}{(\alpha + j\omega)^2}$$

$$E = \frac{1}{R} \int_{-\infty}^{\infty} v^2(t) dt \underset{\text{Parseval}}{=} \frac{1}{R} \int_{-\infty}^{\infty} v(\omega) v^*(\omega) d\omega$$

$$\rightarrow E_{\text{tot}} = \frac{A^2}{R} \int_{-\infty}^{\infty} \frac{1}{(\alpha^2 + \omega^2)^2} d\omega, \quad E_{\omega} = \frac{A^2}{R} \int_{-\omega_{\text{BW}}}^{\omega_{\text{BW}}} \frac{1}{(\alpha^2 + \omega^2)^2} d\omega$$

$$\rightarrow \frac{E_{\omega}}{E_{\text{tot}}} = \frac{\int_{-\omega_{\text{BW}}}^{\omega_{\text{BW}}} \frac{1}{(\alpha^2 + \omega^2)^2} d\omega}{\int_{-\infty}^{\infty} \frac{1}{(\alpha^2 + \omega^2)^2} d\omega} = \frac{\int_0^{\omega_{\text{BW}}} \frac{1}{(\alpha^2 + \omega^2)^2} d\omega}{\frac{\pi}{4\alpha^2}}$$

$$= \gamma \left(\frac{1}{\gamma a^r} \tan^{-1} \frac{\gamma \pi w}{a} + \frac{\gamma \pi w}{a^r (a^r + \gamma \pi^r w^r)} \right) / \frac{\gamma}{\gamma a^r}$$

$$\rightarrow \frac{E_w}{E_{tot}} = \frac{\gamma}{\pi} \left(\tan^{-1} \frac{\gamma \pi w}{a} + \frac{\gamma \pi a w}{a^r + \gamma \pi^r w^r} \right)$$

3.

$$\frac{E_{w, t_c}}{E_{tot}} = \frac{\gamma \int_{w_1}^{w_2} \frac{dw}{(a^r + w^r)^r}}{\int_{-\infty}^{\infty} \frac{dw}{(w^r + a^r)^r}} \quad \text{where: } w_1 = \gamma \pi (t_c - \frac{w}{\pi}),$$

$$w_2 = \gamma \pi (t_c + \frac{w}{\pi})$$

$$= \frac{\gamma}{\pi} \left[\tan^{-1} \frac{w_2}{a} - \tan^{-1} \frac{w_1}{a} + \alpha \left(\frac{w_2}{w_2^r + a^r} - \frac{w_1}{w_1^r + a^r} \right) \right]$$

same as part 2

3 Energy and Power Spectral Density calculation

1. $x_1(t) = e^{-|t|/\alpha} \sin(\beta t)$, x_1 is a damped sine $\rightarrow p=0$
 $\rightarrow$ finite $E \rightarrow$ Energy signal

$$\rightarrow S_x(f) = \mathcal{F} \{ x(t) * x(-t) \} = |X(f)|^2$$

$$x(t) = \frac{\gamma \alpha}{\alpha^r + (\gamma \pi t)^r} * \frac{1}{j\pi} (\delta(t-\beta) - \delta(t+\beta))$$

$$\rightarrow S_x(f) = \left(\frac{E \pi \alpha f \beta f}{(a^r + \gamma \pi (t-f\beta))^r (a^r + \gamma \pi (t+f\beta))^r} \right)^2, \quad f_\beta = \frac{\beta}{\gamma \pi}$$

$$2. x_2(t) = \text{sinc}^2(3t)$$

$$\int_{-\infty}^{\infty} |x_2(t)|^2 dt < \infty \rightarrow x_2 \text{ is an energy signal}$$

$$\rightarrow S_x(f) = F\{x(t) * x^*(-t)\} = |X(f)|^2$$

$$\begin{aligned} X(f) &= F\{\text{sinc}^2(3t)\} = F\{\sin 3t\} * F\{\text{sinc} 3t\} \\ &= \frac{1}{3} \text{rect}\left(\frac{f}{3}\right) * \text{rect}\left(\frac{f}{3}\right) \frac{1}{3} = \frac{1}{9} \Lambda\left(\frac{f}{3}\right) \end{aligned}$$

$$\rightarrow S_x(f) = \frac{1}{9} \Lambda^2\left(\frac{f}{3}\right)$$

$$3. x_3(t) = \sum_{n \in \mathbb{Z}} \Lambda(t - 3n), \quad x = \Lambda(t)$$

$$\begin{aligned} a_k &= \frac{1}{T} F\{x\} \Big|_{f=\frac{k}{T}} = \frac{1}{3} F\{\text{rect}(t) * \text{rect}(t)\} \Big|_{f=\frac{k}{T}} \\ &= \frac{1}{3} F\{\text{rect}(t)\}^2 \Big|_{f=\frac{k}{T}} = \frac{1}{3} \text{sinc}^2\left(\frac{k}{3}\right) \end{aligned}$$

$$x(t) = \frac{1}{3} \sum_{k \in \mathbb{Z}} X\left(\frac{k}{3}\right) e^{j2\pi k \frac{t}{3}}$$

$$\rightarrow S_x(f) = \frac{1}{9} \sum_{k \in \mathbb{Z}} |X\left(\frac{k}{3}\right)|^2 \delta\left(f - \frac{k}{3}\right)$$

$$= \frac{1}{9} \sum_{k \in \mathbb{Z}} \text{sinc}^4\left(\frac{k}{3}\right) \delta\left(f - \frac{k}{3}\right)$$

$$H. x_H(t) = 2u(-t) \quad \int_{-\infty}^{\infty} |x_H(t)|^2 dt = \infty, \quad \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x_H(t)|^2 dt < \infty$$

$\rightarrow x_H$ is a power signal

$$R_x(\tau) = \langle x_H(t) x_H^*(t-\tau) \rangle = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T 2u(-t) u(-t+\tau) dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^0 2u(-t+\tau) dt = \lim_{T \rightarrow \infty} \frac{2(-\tau+T)}{T} = 2u(\tau)$$

$$\rightarrow S_x(f) = \frac{1}{j\pi f}$$

5. $x_5(t) = |\cos(2\pi t)|$: periodic $\rightarrow$ power signal

$$|\cos 2\pi t| = \frac{2}{\pi} + \sum_{k=1}^{\infty} a_k \cos(4\pi k t), a_k = -\frac{4}{\pi(4k^2 - 1)}$$

F.T, $S_x(t) = \frac{4}{\pi^2} \delta(t) + \sum_{k=1}^{\infty} a_k^2 [\delta(t - 2kt) + \delta(t + 2kt)]$

6. $x_6(t) = \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{t^2}{2\sigma^2}}$: Gaussian $\rightarrow$ Energy

$$X(f) = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{t^2}{2\sigma^2}} e^{-j2\pi ft} dt = e^{-2\pi^2 \sigma^2 f^2}$$

$$\rightarrow S_x(f) = |X(f)|^2 = e^{-4\pi^2 \sigma^2 f^2}$$

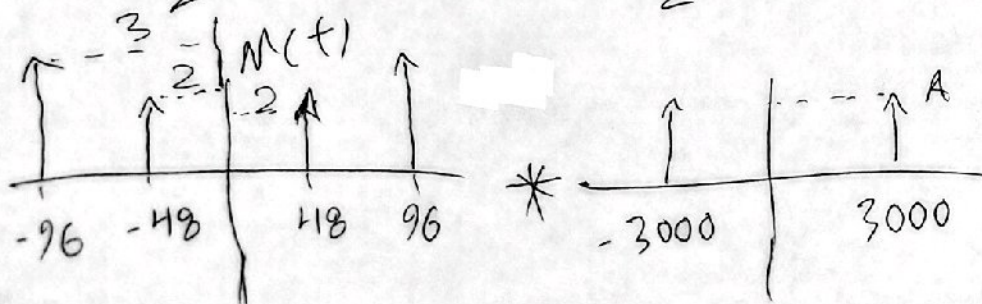
4 DSB

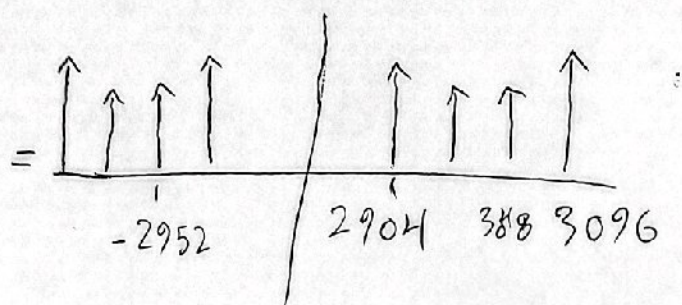
$$s(t) = A \cos(600\pi t) \cdot m(t) = A \cos 600\pi t (2 \cos 300\pi t + 3 \cos 600\pi t)$$

$\cos A \cos B \rightarrow \frac{1}{2} [\cos(A+B) + \cos(A-B)]$

$$s(t) = A \cos(5700\pi t) + A \cos(6300\pi t)$$

$$+ \frac{3A}{2} \cos(5400\pi t) + \frac{3A}{2} \cos(6600\pi t)$$





$$P_s = \frac{1}{2} A^2 P_m, \quad P_m = \frac{2^2}{2} + \frac{3^2}{2} = 6.5 \rightarrow P_s = \frac{13}{4} A^2$$

5 AM

$$1. \text{ carrier} = 20 \cos(2\pi 205t)$$

$$\text{first tone} = \cos(2\pi 2705t) + \cos(2\pi 2295t)$$

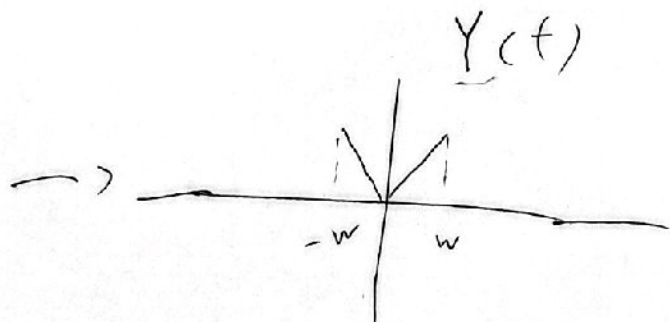
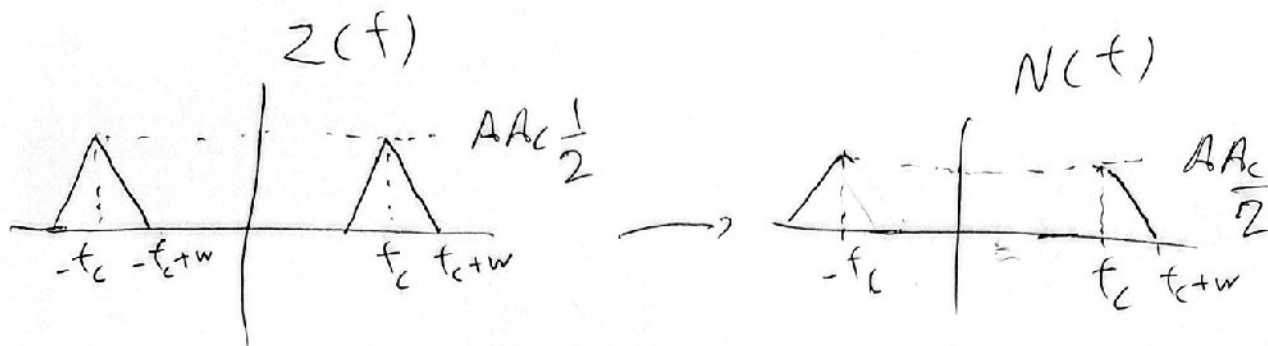
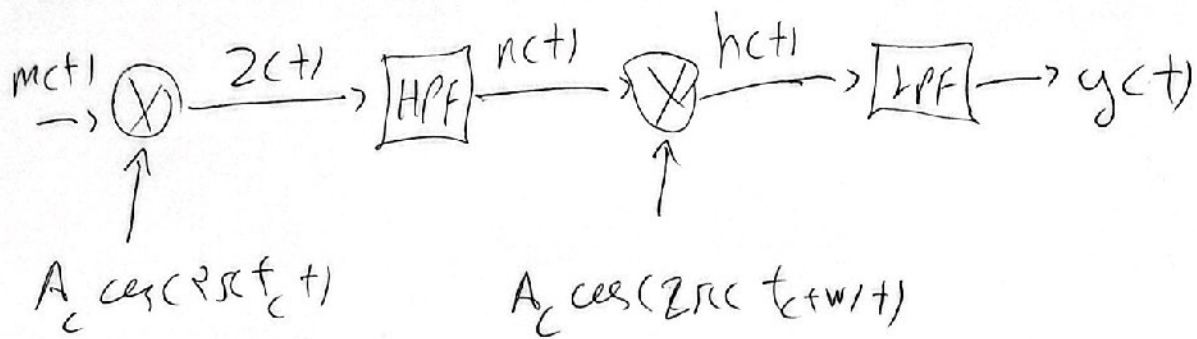
$$\text{sec tone} = 5 \cos(2\pi 5205t) + \cos(2\pi 4795t)$$

$$2. \quad V_{rms} = \frac{A}{\sqrt{2}}, \quad P = \frac{A^2}{R}, \quad P_c = \frac{20^2}{2} = 200 \text{ W}$$

$$P_{LSB1} = \frac{1^2}{2} = 0.5 \text{ W}, \quad P_{LSB2} = \frac{5^2}{2} = 12.5 \text{ W}$$

$$3. \quad u = \frac{A_{max} - A_{min}}{A_{max} + A_{min}} = \frac{32 - 8}{32 + 8} = 0.6$$

$$4. \quad P_{SB, total} = 26 \text{ W}, \quad P_t = 226 \rightarrow \frac{P_{SB}}{P_t} \approx 11.5\%$$



2. $y(t)$ will go through the same process:

